FUYUAN LYU

845 Sherbrooke St W, Montreal, Quebec | fuyuan.lyu@mail.mcgill.ca | fuyuanlyu.github.io

SUMMARY

I'm currently a Ph.D. candidate at McGill University & MILA - Quebec AI Institute, fortunate to be advised by *Prof. Jin L.C. Guo* and *Prof. Xue (Steve) Liu*. I am currently enthusiastic about **extending the boundary of test-time scaling beyond language foundation models**. Correspondingly, I am also building a **modality-native foundation model** in collaboration with other amazing researchers. Specifically, this involves formats such as time series, tabular, and behavioural sequences.

I am currently working as a Quant Researcher at JPMorganChase, aiming to bring my research experience into the finance industry.

Previously, I worked extensively on Data-centric AI, aiming to eventually free humankind from repetitive effort. **I am thrilled that part of my research has successfully transformed into product baselines and made a small difference in the world!** Specifically, my research topics mainly lie in three aspects:

[1] **Advertisement & Personalization**, where I mainly work on fundamental problems such as representation learning [c1, c2, p2], feature selection[c3, c5], feature fusion [c8], label correlation [c6] and their application in real-world scenarios.
[2] **Fintech**, where I collaborate with Tencent Financial Technology on industry-inspired research, including fund allocation [w1, c14, w2], Q&A [c16] and recommendation [c11].
[3] **LLM Evaluation & Reward Model**, which explore the LLM evaluation problem [c7, c10] from various aspects. Together with other amazing collaborators, we also built the world's first taxonomy of test-time scaling [p3].

Before joining McGill, I got my bachelor's degree in Computer Science with *Zhiyuan Honour Degree in Engineering* from Shanghai Jiao Tong University(SJTU). I was advised by *Prof. Li Jiang* for my thesis and worked with *Prof. Xiaokang Yang*. I also had a great time at Nanyang Technological University(NTU), researching with *Prof. Weichen Liu*.

EDUCATION

- PhD, Computer Science, McGill University & MILA - Quebec AI Institute, 2019-2026 (core GPA 3.81/4.00)
- BEng, Computer Science, Shanghai Jiao Tong University, 2015-2019, (core GPA 90.97/100)
- Zhiyuan Honour Degree for BEng, Shanghai Jiao Tong University, 2015-2019

SELECTED PUBLICATIONS

* denotes (co-)first authors, † denotes (co-)corresponding authors.

For the full publication lists, please refer to my GScholar or my DBLP.

Conference:

[c22] HanChong Chen, Jie Liu, **Fuyuan Lyu**, Dugang Liu, Lingjie Li, Xiuqiang He and Xing Tang, "OptST: Optimizing Semantic Tokenization for LLM-Enhanced Recommendation", *In Proceedings of the 20th International Conference on Web Search and Data Mining* (**WSDM 2027 Findings**)

[c21] Xing Tang, **Fuyuan Lyu**, Shaoxiang Qin, Chen Ma and Xiuqiang He, "One Value, Multiple Views: Composing Numerical Representations for Click-Through Rate Prediction", *In Proceedings of the 20th International Conference on Web Search and Data Mining* (**WSDM 2027**)

[c20] Shaoxiang Qin, Yucheng Zhao, **Fuyuan Lyu**, Di Zhou, Jiachen Yao, Xue Liu, Anima Anandkumar, Liangzhu Leon Wang and Xiongye Xiao, "GeoWind2Plan: Mission-Time 3D Urban Wind Prediction for Energy-Efficient UAV Planning", *In Proceedings of the Fortieth Annual Conference on Neural Information Processing Systems* (**NeurIPS 2026 spotlight**)

[c19] **Fuyuan Lyu**, Zhentai Chen, Jingyan Jiang, Lingjie Li, Xing Tang, Xiuqiang He, and Xue Liu, "Exploring Test-time Scaling via Prediction Merging on Large-Scale Recommendation", *In Proceedings of the 49th International ACM SIGIR Conference on Research and Development in Information Retrieval* (**SIGIR 2026**)

[c18] Qiyuan Zhang, Junyi Zhou, Yufei Wang, **Fuyuan Lyu**, Yidong Ming, Can Xu, Qingfeng Sun, Kai Zheng, Peng Kang, Xue Liu, and Chen Ma, "RubricBench: Aligning Model-Generated Rubrics with Human Standards", *In Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics* (**ACL 2026**)

[c17] Linfeng Du, Yuan Ye, Zichen Zhao, **Fuyuan Lyu**, Emiliano Penaloza, Xiuying Chen, Zipeng Sun, Jikun Kang, Laurent Charlin, Xue Liu, and Haolun Wu, "Optimizing User Profiles via Contextual Bandits for Retrieval-Augmented LLM

Personalization", *In Proceedings of The 64th Annual Meeting of the Association for Computational Linguistics* (**ACL 2026**)

[c16] Xing Tang, Hao Chen, Shiwei Li, **Fuyuan Lyu**, Weijie Shi, Lingjie Li, Dugang Liu, Weihong Luo, Xiku Du, and Xiuqiang He, "Data-Driven Function Calling Improvements in Large Language Model for Online Financial QA", *In Proceedings of the Web Conference 2026, Industrial Track* (**WWW 2026**)

[c15] Xing Tang, Chaohua Yang, Yuwen Fu, Dongyang Ao, Shiwei Li, **Fuyuan Lyu**, Linfeng Du, and Xiuqiang He, "Retrieval-Augmented Cross-Domain LifeLong Behaviour Modelling for Enhancing Click-through Rate Prediction", *In Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining, Applied Data Science Track* (**KDD 2025** 2nd round)

[c14] **Fuyuan Lyu**, Linfeng Du, Yunpeng Weng, Qiufang Ying, Zhiyan Xu, Wen Zou, Haolun Wu, Xiuqiang He, and Xing Tang, "Timing is important: Risk-aware Fund Allocation based on Time-Series Forecasting", *In Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining, Applied Data Science Track* (**KDD 2025** 2nd round)

[c13] Qiyuan Zhang, Yufei Wang, Yuxin Jiang, Liangyou Li, Chuhan Wu, Yasheng Wang, Xin Jiang, Lifeng Shang, Ruiming Tang, **Fuyuan Lyu**, and Chen Ma, "Crowd Comparative Reasoning: Unlocking Comprehensive Evaluations for LLM-as-a-Judge", *In Proceedings of The 63rd Annual Meeting of the Association for Computational Linguistics* (**ACL 2025**)

[c12] Ziqiang Cui, Yunpeng Weng, Xing Tang, **Fuyuan Lyu**, Dugang Liu, Xiuqiang He, and Chen Ma, "CoLaKG: Comprehending Knowledge Graphs with Large Language Models for Recommender Systems", *In Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval* (**SIGIR 2025**)

[c11] Xing Tang*, Yunpeng Weng*, **Fuyuan Lyu***, Dugang Liu, and Xiuqiang He, "A Predict-Then-Optimize Customer Allocation Framework for Online Fund Recommendation", *In Proceedings of The 30th International Conference on Database Systems for Advanced Applications* (**DASFAA 2025**)

[c10] Qiyuan Zhang, Yufei Wang, Tiezheng Yu, Yuxin Jiang, Chuhan Wu, Liangyou Li, Yasheng Wang, Xin Jiang, Lifeng Shang, Ruiming Tang, **Fuyuan Lyu**, and Chen Ma, "RevisEval: Improving LLM-as-a-Judge via Response-Adapted References", *In Proceedings of the Thirteenth International Conference on Learning Representations* (**ICLR 2025**)

[c9] Dugang Liu, Chaohua Yang, Yuwen Fu, Xing Tang, Gongfu Li, **Fuyuan Lyu**, Xiuqiang He, and Zhong Ming, "Scenario Shared Instance Modeling for Click-through Rate Prediction", *In Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining, Research Track* (**KDD 2025** 1st round)

[c8] Kexin Zhang*, **Fuyuan Lyu***, Xing Tang, Dugang Liu, Chen Ma, Kaize Ding, Xiuqiang He, and Xue Liu, "Fusion Matters: Learning Fusion in Deep Click-through Rate Prediction Models", *In Proceedings of the 18th International Conference on Web Search and Data Mining* (**WSDM 2025**)

[c7] Qiyuan Zhang, **Fuyuan Lyu**†, Xue Liu, Chen Ma†, "Collaborative Performance Prediction for Large Language Models", *In Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing* (**EMNLP 2024**)

[c6] Xing Tang*, Qiao Yang*, **Fuyuan Lyu***, Dugang Liu, Xiuqiang He, "Touch the Core: Exploring Task Dependence Among Hybrid Targets for Recommendation", *In Proceedings of the 18th ACM Conference on Recommender Systems* (**Recsys 2024**)

[c5] **Fuyuan Lyu**, Xing Tang, Dugang Liu, Chen Ma, Weilong Luo, Xiuqiang He, and Xue Liu, "Towards Hybrid-grained Feature Interaction Selection for Deep Sparse Network", *In Proceedings of the 37th Conference on Neural Information Processing Systems* (**NeurIPS 2023**)

[c4] Dugang Liu, Xing Tang, Han Gao, **Fuyuan Lyu**, and Xiuqiang He, "Explicit Feature Interaction-aware Uplift Network for Online Marketing", *In Proceedings of The 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, Applied Data Science Track* (**KDD 2023**)

[c3] **Fuyuan Lyu***, Xing Tang*, Dugang Liu, Chen Liang, Xiuqiang He, and Xue Liu, "Optimizing Feature Set for Click-through Rate Prediction", *In Proceedings of The Web Conference 2023* (**WWW 2023**)

[c2] **Fuyuan Lyu***, Xing Tang*, Hong Zhu, Huifeng Guo, Yingxue Zhang, Ruiming Tang, and Xue Liu, "OptEmbed: Learning Optimal Embedding Table for Click-through Rate Prediction", *In Proceedings of the 31st ACM International Conference on Information and Knowledge* (**CIKM 2022**)

[c1] **Fuyuan Lyu***, Xing Tang*, Huifeng Guo, Ruiming Tang, Xiuqiang He, Rui Zhang, and Xue Liu, "Memorize, Factorize, or be Naïve: Learning Optimal Feature Interaction Methods for CTR Prediction", *In Proceedings of the 38th IEEE International Conference on Data Engineering* (**ICDE 2022**)

Journal:

[j1] Haolun Wu, Yansen Zhang, Chen Ma, **Fuyuan Lyu**, Bowei He, Bhaskar Mitra, and Xue Liu, "Result Diversification in Search and Recommendation: A Survey", *IEEE Transactions on Knowledge and Data Engineering (2024)*

Workshop:

[w4] Shaoxiang Qin, **Fuyuan Lyu**, Di Zhou, Xue Liu, Xiongye Xiao, Anima Anandkumar, and Liangzhu Wang, "From Large-Scale Winds to Urban Decision Making: A Cross-Scale Framework for Wind-Aware UAV Navigation", *ICLR 2026 Workshop on AI and Partial Differential Equations*

[w3] Zhengxi Li, **Fuyuan Lyu**, Qiyuan Zhang, Ye Yuan, Haolun Wu, and Xue Liu "Towards Reasoning Reuse: A New Paradigm in Model Collaboration", *ICLR 2026 Workshop on Test-Time Updates*

[w2] Rassul Magauin, **Fuyuan Lyu**, Lu Han, and Xue Liu, "Continuous RTS-PnO: Constraint-Aware Training for Robust Rolling-Horizon Budget Allocation", *ICLR 2026 Workshop on AI for Mechanism Design and Strategic Decision Making*

[w1] **Fuyuan Lyu**, Linfeng Du, Yunpeng Weng, Qiufang Ying, Zhiyan Xu, Wen Zou, Haolun Wu, Xiuqiang He, and Xing Tang, "Timing is important: Risk-aware Fund Allocation based on Time-Series Forecasting", *ICLR 2025 Workshop on Advances in Financial AI: Opportunities, Innovations and Responsible AI*

Preprint:

[p5] **Fuyuan Lyu**, Chenglin Luo, Qiyuan Zhang, Yupeng Hou, Haolun Wu, Xing Tang, Xue Liu, Jin L.C. Guo, and Xiuqiang He, "Give Users the Wheel: Towards Promptable Recommendation Paradigm", *arxiv 2602.18929*

[p4] Dun Yuan*, **Fuyuan Lyu***, Ye Yuan, Weixu Zhang, Bowei He, Jiayi Geng, Linfeng Du, Zipeng Sun, Yankai Chen, Changjiang Han, and Jikun Kang, Alex Chen, Haolun Wu, and Xue Liu "Beyond Message Passing: A Semantic View of Agent Communication Protocols", *arxiv 2604.02369*

[p3] Qiyuan Zhang*, **Fuyuan Lyu***, Zexu Sun, Lei Wang, Weixu Zhang, Zhihan Guo, Yufei Wang, Irwin King, Xue Liu, and Chen Ma, "What, How, Where, and How Well? A Survey on Test-Time Scaling in Large Language Models", *arxiv 2503.24235*

[p2] **Fuyuan Lyu**, Xing Tang, Dugang Liu, Haolun Wu, Chen Ma, Xiuqiang He, and Xue Liu, "Feature Representation Learning for Click-through Rate Prediction: A Review and New Perspectives", *arxiv 2302.02241*

[p1] Yuecai Zhu, **Fuyuan Lyu**, Chengming Hu, Xi Chen, and Xue Liu, "Encoder-Decoder Architecture for Supervised Dynamic Graph Learning: A Survey", *arxiv 2203.10480*

EXPERIENCE

JP Morgan Chase **Oct 2026 to now**

Quant Researcher

- TBD

Vatic Labs **Jan 2026 to Sep 2026**

Quant Researcher

- Developed and co-managed a minute-level statistical arbitrage pipeline, leveraging sequential modeling to capture non-linear alphas on CME Futures.
- Developed LOB feature extraction and representation, processing L3 order book data to generate high-fidelity predictive signals.
- Initiated a rigorous alpha evaluation framework, utilizing robust out-of-sample backtesting to measure strategy capacity, IC, capture ratio, and monetizability.

McGill University & MILA **Sep 2019 to now**

Research Assistant *Advisor: Prof. Jin L.C. Guo & Prof. Xue Liu*

- OptFusion: This paper investigates the fusion mechanism in the CTR model. Specifically, we found that the embedding layer is the juicy part of the model. In order to perform well, a model does not need to be heavy but needs to be densely connected to the embedding layer. The paper is accepted at WSDM 2025.

- OptNum: This paper studies the numerical modelling in tabular prediction tasks. The paper is under submission to TMLR.
- OptFeature: Propose to select feature interactions at different granularities in deep sparse networks. The paper is accepted at NeurIPS 2023.
- OptFS: Jointly consider the influence of selecting features and feature interactions in CTR Prediction and propose an automated method to select. The paper is accepted at WWW 2023. It has been utilized on the Tencent FiT platform with millions of annual revenue increases.

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI) **Sep 2025 to Dec 2025**
Visiting Student *Host: Prof. Xue Liu*
- Test-time Scaling on Modality-native Foundational Models: This paper explores test-time scaling via prediction merging on large-scale recommendation and demonstrates that it has better scaling ability than naively expanding the model size. The paper is accepted at SIGIR 2026.
- Give Users the Wheel: Towards Promptable Recommendation Paradigm: This paper explores a new recommendation paradigm where the user can directly prompt an existing recommendation model.

RBC Borealis **Jan 2025 to Jun 2025**
Research Intern *Advisor: Dr. Melissa Mozifian & Dr. Ankit Vani & Dr. Edward Smith*
- RL4ExplainableTS: Utilizing Reinforcement Learning for explainable time series forecasting. This project is under submission to TMLR.

City University of Hong Kong **Mar 2024 to Sep 2024**
Research Assistant *Advisor: Prof. Chen Ma*
- CPP4LLM-Eval: We propose a Collaborative Performance Prediction framework for LLM Evaluation. This project revisits the scaling law from the collaborative perspective. This paper has been accepted by EMNLP 2024.
- RevisEval: This project aims to better utilize the LLM's generative ability instead of its discriminative ability in LLM-as-a-Judge. Specifically, the quality of an answer is not directly determined by asking the LLM to give it a score. Instead, it is measured as the distance between the original answer and the answer revised by the LLM. This paper has been accepted by ICLR 2025.

Tencent, FiT **Jan 2024 to Sep 2024**
Research Intern *Advisor: Dr. Xing Tang & Dr. Xiuqiang He*
- RTS-PnO: This paper is inspired by a product-line problem: how to allocate a given amount of assets over a period of time in the Tencent's FiT department. A predict-then-optimize solution extended over the time domain is proposed as a practical solution. This paper is accepted by KDD ADS 2025.
- PTOFA: This paper is inspired by a product-line problem: fund recommendation in Tencent's FiT department. Unlike common recommendations, different funds have a strict minimum exposure to users, and the revenue gaps between users are much more significant. This paper is accepted by DASFAA 2025.
- HTLNet: This paper is inspired by a product-line problem: customer revenue estimation under certain investments in Tencent's FiT department. A hybrid-target learning network is proposed as a practical solution. This paper was accepted by RecSys 2024.

Huawei Noah's Ark Lab, Montreal **Nov 2021 to Jun 2023**
Research Intern *Advisor: Yingxue Zhang*
- Towards Synthetic Negative Sample Generation for Implicit Recommendation: In this project, we challenge the common belief in the negative sampling community that a "best" negative sampler exists. Instead, we hypothesize that the negative sampler is model&data-dependent and propose a synthetic sample generation algorithm.
- OptEmbed: Study the influence of the size of the embedding table in CTR Prediction. Propose a unified framework to optimize the size of the embedding table and model parameters automatically. The paper is accepted in CIKM 2022.

Huawei Noah's Ark Lab **Mar 2021 to Aug 2021**
Research Intern *Advisor: Dr. Xing Tang & Dr. Ruiming Tang*
- OptInter: Study the influence of modelling methods for feature interaction. Propose a novel deep CTR prediction framework including various modelling methods and a two-stage learning algorithm to select the optimal modelling methods for each feature interaction automatically. The paper is accepted in ICDE 2022.

Shanghai Jiao Tong University **Feb 2019 to Aug 2019**
Research Assistant *Advisor: Prof. Li Jiang*

- Dual-flow Training Framework to Exploit Structured Sparsity: Support structured sparsity from framework level. Propose a flexible dual-flow mechanism to decouple the non-zero data and the sparse network structure. Integrate the dual-flow mechanism into the deep learning compiler stack built on TVM.

Nanyang Technological University **Aug 2018 to Feb 2019**
Research Assistant *Advisor: Prof. Weichen Liu*

- Cross-filter Compression: Address the conflict between uniform quantization and compact network design when speeding up CNN. Explored CNN spatial-adjunct layer property and proposed cross-filter compression method.

Shanghai Jiao Tong University **Jun 2017 to Feb 2018**
Research Assistant *Advisor: Prof. Xiaokang Yang*

- Explored the possibility of more realistic cloth transfer by generating and transferring structures in 3D.
- Designed a specific generative adversarial network to animate cartoon figures.

TEACHING AND MENTORING EXPERIENCE

- Teaching
  - Teaching Assistant for COMP303: *Software Design*, 2020 Winter
- Mentoring:
  - Helen Zhao on Classical ML, 2025 Winter
  - Dowoo Kim on Uncertainty for Treatment Effect Estimation, 2025 Winter
  - Carlone Scott on Automatic Data Labelling, 2024 Summer & Fall
  - Qiyuan Zhang on LLM Evaluation, 2024 Winter & Summer
  - Kexin Zhang(now RA@UCSD) on Recommender System, 2024 Winter
  - Tim Yang & Tom Yang on LLM-based Feature Selection, 2024 Winter
  - Aasheesh Singh(now MLE@Vector) on Federated Prediction, 2023 Summer & Fall
  - Shaoxiang Qin on Fourier Neural Operator, 2023-2024

SELECTED AWARDS

- National Award for Outstanding Overseas Student (Class A), 2023
- NeurIPS Scholar Award, 2023
- McGill GREAT Award for NeurIPS, 2023
- SIGIR Student Travel Grant for CIKM, 2022
- Grad Excellence Award of McGill University, 2019-2024
- Faculty of Science Graduate Supplement Award, 2022-2024
- Natural Sciences and Engineering Research Council of Canada, 2019-2020 & 2022-2024
- School of Computer Science Grad Stimulus Initiative Award, 2020-2021
- Wolfe Fellowship in Sci & Tech, 2019-2020
- Zhiyuan Honor Degree of Engineering of Shanghai Jiao Tong University(1%), 2019
- Zhiyuan Oversea Research Scholarship (1st class) of Shanghai Jiao Tong University, 2019
- Zhiyuan Outstanding Student Scholarship of Shanghai Jiao Tong University(5%), 2019
- Microsoft Intelligence Award , awarded to 4/438 participants in the 1st HACKxFDU Hackathon, 2016.